Architectural Justification: Telemedicine Slot Booking & Prescription Portal

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Architecture Selection: *"We chose Layered Architecture for the Telemedicine Slot Booking & Prescription Portal System."*

The system is organised as a Presentation Layer (Patient Portal UI, Physician Portal UI), a Business Layer (Booking Manager, Consultation Room Service, Prescription Service) and a Data & Integration Layer (Database, Notification Service, Audit Logger, external License Validation). Components talk only through the interfaces in the diagram: IBooking, IRoomLink, IPrescriptionAccess, IPrescriptionAuthoring, INotify, IDataAccess, IAuditLog and ILicenseValidation.

Style analysis (Step 2)

Style	Fit for this portal	Main drawback here
Layered	Matches the clear UI → booking/prescription logic → data split; easy to secure and test layer by layer.	Every request passes through each layer; the whole app scales as one unit.
Microservices	Booking and prescriptions could scale and deploy independently.	Network calls, distributed data consistency (double-booking) and ops cost are excessive for 5 core stories.
Client-Server	Simple: browser clients and one central server.	No internal structure; booking, signing and auditing logic would be tangled in one server tier.

Two scenario-specific reasons

1. **The workflow is naturally layered and dependency-ordered.** A patient views doctors, books a slot (FR-001/002), then receives a room link (FR-003); a physician signs a prescription only after license validation (FR-004) before the patient downloads it (FR-005). Each step is a business rule sitting between a UI and stored data, so the UI never touches the database directly and the rules (e.g. "an already-booked slot cannot be confirmed again") live in one place.
2. **One consistent data layer prevents double-booking and keeps records verifiable.** Slot reservation needs a single transactional check-and-reserve on shared data. A layered design keeps that behind one IDataAccess interface, avoiding the distributed-consistency problems microservices would bring, while the team (Sprint 1, 19 points) can still build and test each layer separately.

Security advantage

The layers act as defence in depth for patient and prescription data (NFR-002). Clients can reach only the Business Layer's HTTPS interfaces, where authentication and authorization are enforced (only the authorized patient receives the room link or the signed PDF). Only the Business Layer can call IDataAccess (TLS in transit, AES-256 at rest), and every booking, room-access and signing event is written through IAuditLog, giving one auditable trail. The external license registry is isolated behind ILicenseValidation, so signing is blocked if validation fails.

Performance benefit

Because the Booking Manager talks to the database through a single local interface rather than a chain of network hops between services, a booking request needs only one client round-trip and one database transaction, which supports the 2-second booking-response target (NFR-001). Slow side-tasks (sending notifications, writing audit logs) are handled asynchronously by the lower layer, so they do not delay the patient's confirmation; the business tier is stateless and can be replicated behind a load balancer to support 99.9% availability.

Telemedicine Slot Booking & Prescription Portal — UML Component Diagram (Layered Architecture)

